

**The Moon's Rarest Lava: Tracking Silica-Rich Volcanism Through Gruithuisen and Mairan's
Slopes and Christiansen Feature Signatures**

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Abstract

Silicic volcanism is rare on the Moon relative to the extensive basaltic maria, making silicic domes unusually diagnostic features for evaluating how evolved melts are generated, emplaced, and modified on an airless body. Since lunar fieldwork is limited, remote sensing provides the primary means to characterize these constructs by mapping their topographic expression and mineralogical proxies consistently across broad areas. In this study, we combine laser altimetry-derived topography with thermal-infrared spectroscopy to compare two prominent silicic provinces in northern Oceanus Procellarum: the Gruithuisen volcanic complex (specifically Gamma and Delta domes) and the Mairan domes. Gridded Lunar Orbiter Laser Altimeter (LOLA) digital elevation models obtained from the PDS Geosciences Node were processed in ENVI 6.1 to generate shaded-relief and slope-in-degrees products, and matched regions of interest (domes and adjacent control terrains) were used to quantify slope distributions and slope-area fractions across summit, flank, and margin zones. To constrain composition, LRO Diviner observations of the Christiansen Feature (CF) were evaluated alongside CF-FeO deviations from local baselines as an orbital indicator of silica enrichment within the same ROIs. This integrated framework tests whether apparent silica-rich signatures preferentially occur on specific morphological settings (e.g., summit caps versus steepened margins) and provides a reproducible approach for comparing the topography and compositional driven signals across the Gruithuisen and Mairan dome systems. Results suggest silicic anomalies relate to steep slopes and non-uniform dome morphology, similar to characteristics of terrestrial rhyolitic volcanic features. High viscosity composite flows from increased silica proportions drive these similarities, however, an absence of water and plate tectonics on the Moon make the origins of the Gruithuisen and Mairan domes unclear.

Introduction

Occurring in a setting where the maria typically smooth and level regional lows, the Gruithuisen and Mairan domes are the focuses of our topographical and compositional analysis: Gruithuisen domes are centered near 36.3°N, 319.8°E, where orbital imaging highlights an isolated, high-standing

volcanic complex surrounded by mare lavas. The Mairan system occurs nearby at 43.673°N, 310.145°E.

Morphologically, lunar domes are typically low-relief, constructional edifices with convex profiles and generally gentle slopes, but the Gruithuisen and Mairan domes display pronounced margins and locally steepened flanks that may reflect higher-viscosity emplacement and/or subsequent degradation. Compositionally, LRO Diviner thermal-infrared measurements constrain the Christiansen Feature (CF), which shifts to shorter wavelengths with increasing silicate polymerization. Deviations in the expected silica-albedo relationship provide an orbital proxy for identifying silicic anomalies, quantifiable by comparing CF placement to local terrain baselines.

The lunar crust is abundant in the silicate minerals plagioclase feldspar, pyroxene, and olivine across both the anorthitic highlands and basaltic mare. In the thermal infrared, silicate mineral spectra exhibit a Christiansen Feature—a reflectance minimum, emission maximum—which shifts to shorter wavelengths with increasing degrees of silicate polymerization due to the shortening and strengthening of bonds. Of the abundant lunar silicates, Olivine has the lowest degree of polymerization. The oxygen atoms of the isolated silicate tetrahedron are not shared with other tetrahedra so it has the lowest bond energy and longest CF wavelength at 8.47 μ m. Pyroxene, a chain silicate, has an intermediate degree of polymerization. It shares two oxygen atoms with other tetrahedra and has a CF wavelength of 8.13 μ m. Anorthite feldspar, a calcium-rich plagioclase, is a framework silicate representing the highest degree of polymerization. It shares all four oxygen atoms with other tetrahedra so it has the highest bond energy and shortest CF wavelength at 7.81 μ m (Lucey et al. 2021). Our observation of CF placement in thermal infrared data will inform our compositional comparison of lunar surface features.

To investigate the variations in slope of the domes, we analyzed slope statistics using the LOLA–SELENE Digital Elevation Model (SLDEM) obtained from the PDS Geosciences Node Orbital Data Explorer (ODE). SLDEM integrates LOLA laser altimetry with SELENE (Kaguya) Terrain Camera stereo-derived topography to generate a near-global lunar DEM. Specifically, we used matched dome and control-terrain regions of interest to investigate how dome-scale morphology covaries with apparent silica-rich composition across each dome system and relative to adjacent

terrains, and what consistent or divergent patterns imply about the correlation between composition and slope about the silicate domes.

Methodology

Silica Variance

Lucey et al. (2021) used three narrow-band IR channels and calibrated radiances from the Planetary Data System archive to derive global CF values corrected for illumination, topography, and space weathering. However, the proportion of mineral composition cannot be determined from CF position alone. A $8.13\mu\text{m}$ CF can be diagnostic of pure pyroxene but since it falls between $\text{CF}_{\text{Olivine}}$ and $\text{CF}_{\text{Anorthite}}$, it might instead be a mixture of olivine and anorthite or some mixture of all three minerals (Lucey et al. 2021). To clarify this ambiguity, albedo can be used to help distinguish between the endmember species due to its correlation with mineral composition. This relationship can be observed when comparing the reflectance of lunar highlands dominated by felsic minerals like anorthite feldspar with the lunar maria which are dominated by mafic minerals like olivine, pyroxene, and ilmenite.

However, this correlation between albedo and mineral content breaks down for silica-rich and olivine-rich locations which do not exhibit extreme albedo but have shifted CF wavelengths due to differing TIR reflectance of the bulk mixture. As a result, silicic and ultramafic compositions deviate from the CF-albedo relationship expected from the typical plagioclase–pyroxene–olivine mixture of the lunar surface. However, albedo is a reflectance function of factors including illumination, solar weathering, and slope. To minimize these effects, an NIR-derived FeO accounting for these factors is used as a proxy due to its anticorrelation with albedo as a strongly absorbent iron oxide. Using a FeO to create a 1:1 correlation with CF (scaled to properly represent the compositional differences between the highlands and maria), we can detect silicic and ultramafic regions from their respective negative and positive anomalies (Lucey et al 2021). (Fig 1) A negative deviation signifies a shift of the CF to a shorter wavelength without changing the albedo, such as from the presence of a highly polymerized silica among a bright plagioclase mixture. Conversely, an olivine-rich location—silica poor and iron

enriched—will produce a positive anomaly because the CF will shift to longer wavelengths due to the depletion of silica while the bulk mixture exhibits an albedo resembling the dark mafic basalts.

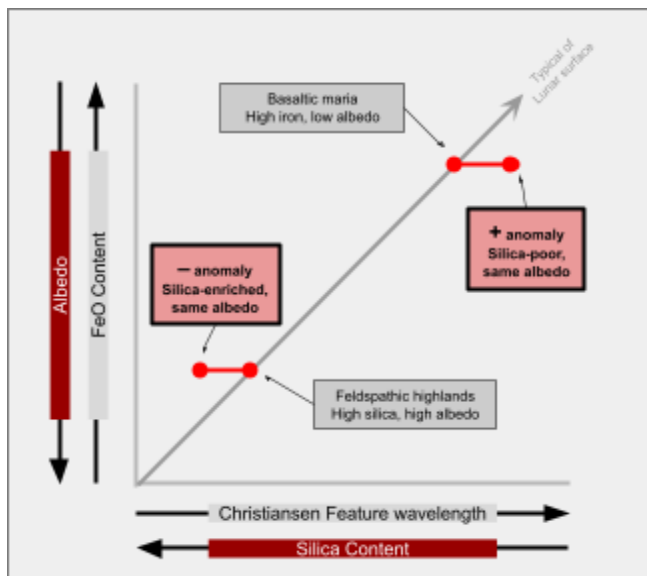


Fig. 1: Surface material representative of typical lunar mineralogy falls along the 1:1 correlation between CF wavelength and FeO content. Silicic and ultramafic anomalies fall off of this line because the proportion of silica shifts the CF position without changing the albedo of the bulk mixture.

Using the map of CF-FeO correlation, we verified the presence of atypical silicic compositions across the two Gruithuisen domes on the western rim of the Ibrum basin and four Mairan domes on the Northern Oceanus Procellarum by their negative anomalies. We used a CF map to quantify the resulting changes in the TIR spectra (Greenhagen et al. 2021). To determine the dome boundaries, we used the Global Lunar DTM (GLD100) shown in Figure 2 to trace positive relief in order to minimize the topographic uncertainty caused by illumination (Scholten et al. 2010). In order to accurately compare map features, the global CF and CF-FeO maps were subset to match the extent of the DTM, which covers a latitude of 0°N to 60°N and longitude from 0°W to 90°W, then scaled to the same dimensions. In addition to the domes, we calculated statistics for a topographically flat and spectrally uniform highland and mare region near each site as a baseline comparison.

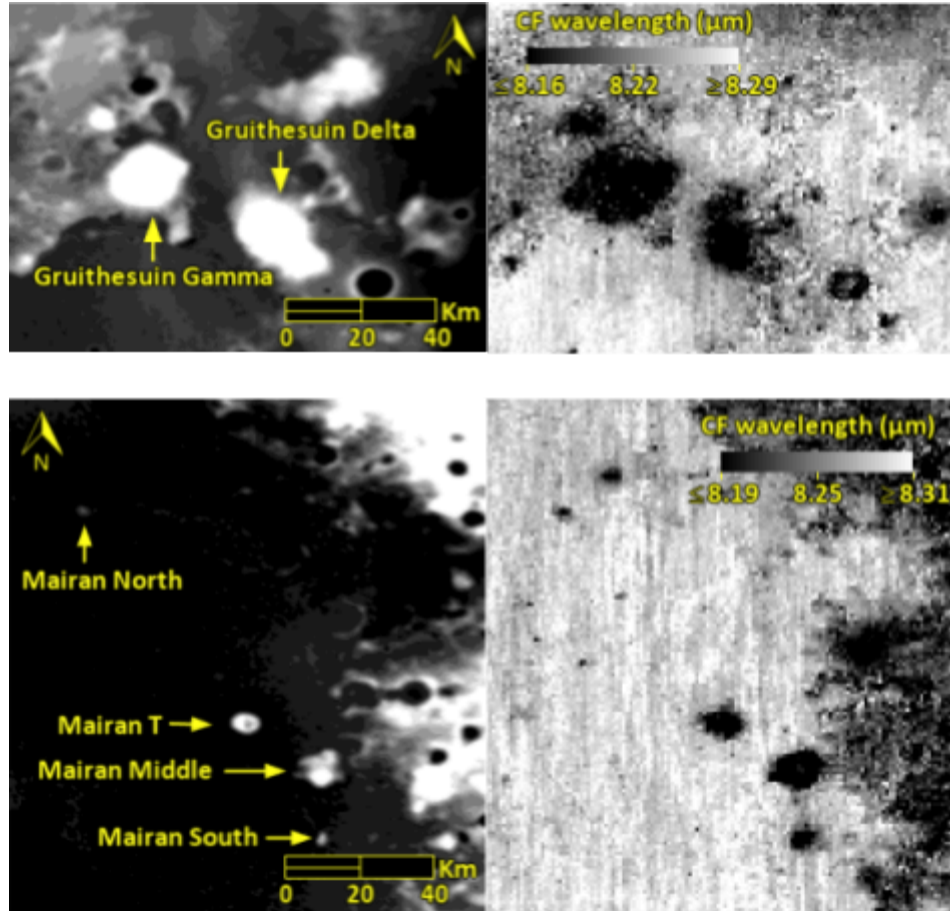


Fig. 2: DTM (left) and CF (right) maps of study sites. Regions of interest used to compute summary statistics for the domes were created using the DTM to more confidently trace positive topographic relief.

Slope Morphology

In SLDEM, each raster pixel value represents surface elevation (meters) relative to the standard lunar reference sphere with a radius of 1737.4 km, providing a consistent vertical datum for comparing relief across domes and adjacent terrains (Neumann, 2010). The dataset has a horizontal sampling of 512 pixels per degree and typical vertical accuracy of ~3–4 m, enabling robust quantification of dome-scale topographic gradients and breaks in slope. First, we established consistent regions of interest (ROIs) on the SLDEM surfaces in which every statistic—whether slope histograms or later comparisons to Diviner CF—was computed over the same, explicitly defined processes. To guide ROI

placement, we generated hillshade images with a 3×3 kernel, azimuth of 77°, and illumination elevation of 40°, producing an oblique lighting geometry that enhances breaks in slope around dome margins while retaining internal texture (Fig.3).

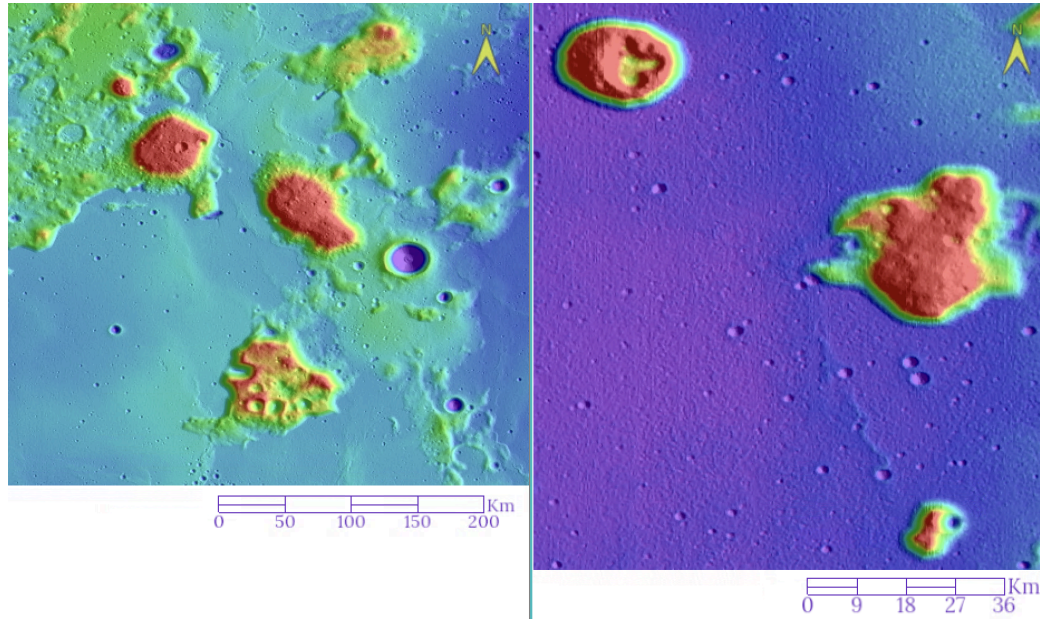


Fig. 3: Hillshade images of the Gruithuisens (left) and Mairans (right). Note that red represents the highest elevation and blue indicating the lowest.

These two gradient components are combined to produce a single steepness magnitude, which is reported as an angle in degrees ranging from near-zero on flat surfaces to high values on steep faces. This slope raster is the primary metric used here to diagnose constructional relief: it delineates the transition from relatively smooth summit caps to steeper flanks and, ultimately, to the low-slope mare plains (Fig. 4). In the topography-derived slope visualization, brighter annular belts commonly correspond to steeper dome margins and flank segments, whereas darker interiors tend to represent gentler cap surfaces.

As both the Gruithuisen and Mairan provinces exhibit heterogeneous morphologies and complex volcanic–impact overprinting, we summarize each target by computing minimum, mean, and maximum slope values within ROIs defined for individual domes and adjacent control terrain. This approach is motivated by prior morphometric work that shows steepness is not uniformly distributed

across silicic domes and that slope breaks can mark meaningful structural boundaries. For instance, Ivanov et al. (2016) report that Gruithuisen Gamma includes a broad, gently sloping summit plateau bounded by pronounced slope breaks and scalloped scarps, and they interpret these sharp transitions—together with crater-exposed layering—as evidence for construction by high-viscosity effusive emplacement with spatially variable internal facies. In our analysis, this motivates separating summit, flank, and margin domains in the ROI design and using slope extremes (not only mean values) to capture narrow but diagnostic steep belts that may correspond to scarps, lobe margins, or constructional break-in-slope surfaces.

At Mairan, Boyce et al. (2018) interpret the “middle dome” as a composite silicic complex composed of multiple small edifices formed during more than one emplacement episode, with subsequent modification by impacts and mare embayment. On top of that, we treat Mairan as a set of sub-units (e.g., Mairan T, middle, south, and nearby cratered terrain) rather than a single homogeneous dome and explicitly comparing dome ROIs to surrounding mare controls. In this framework, slope distributions are especially diagnostic: when steep slopes appear as extended high-slope tails confined to specific sectors instead of being evenly expressed around the entire perimeter, they are consistent with incremental, multi-phase dome construction that can generate overlapping lobes and internal breaks-in-slope (Carrasco-Núñez and Riggs, 2008). Eventually, these literature constraints motivate our utilization of slope statistics as a quantitative test of whether steepness patterns primarily record constructional morphology (e.g., discrete margin belts and lobe-like flanks) or are influenced by later modifications (e.g., embayed, low-slope toes).

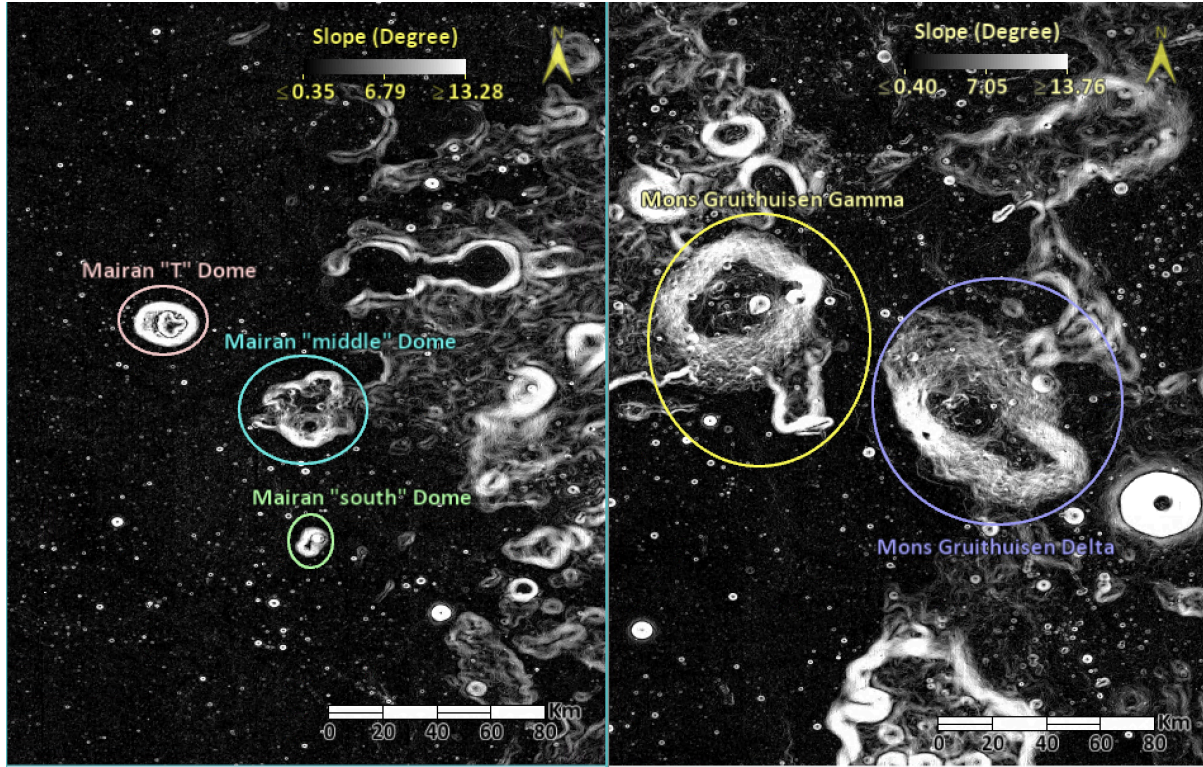


Fig. 4 (Left): The Topography-Derived Slope Map of Marian T, Marian “middle,” and Marian “south” domes. Center of the image at 41.77°N and 318.96°E. **Fig 4. (Right):** The Topography-Derived Slope Map of Mons Gruithuisen Gamma (with mean diameter around 22km) and Delta (with mean diameter around 25km). Center of the Image at 36.37°N and 319.67°E.

Accordingly, beyond the whole-ROI mean and maximum slope, we evaluated an average flank slope by extracting radial slope profiles and averaging only the flank segment (i.e. operationally as the outer sloping zone between the margin and the toe where slopes relax onto the mare control surface). This provides a comparison between overall mean slope, flank mean slope, and flank maxima, addressing whether each dome’s steepness is dominated by narrow margin belts or by broadly steep flanks (fig. 5). Interpreting steepened dome margins as potential expressions of evolved, higher-viscosity emplacement within a region otherwise dominated by low-viscosity flows is consistent with bimodal volcanic framework documented in terrestrial settings (Brewer et al., 2004).

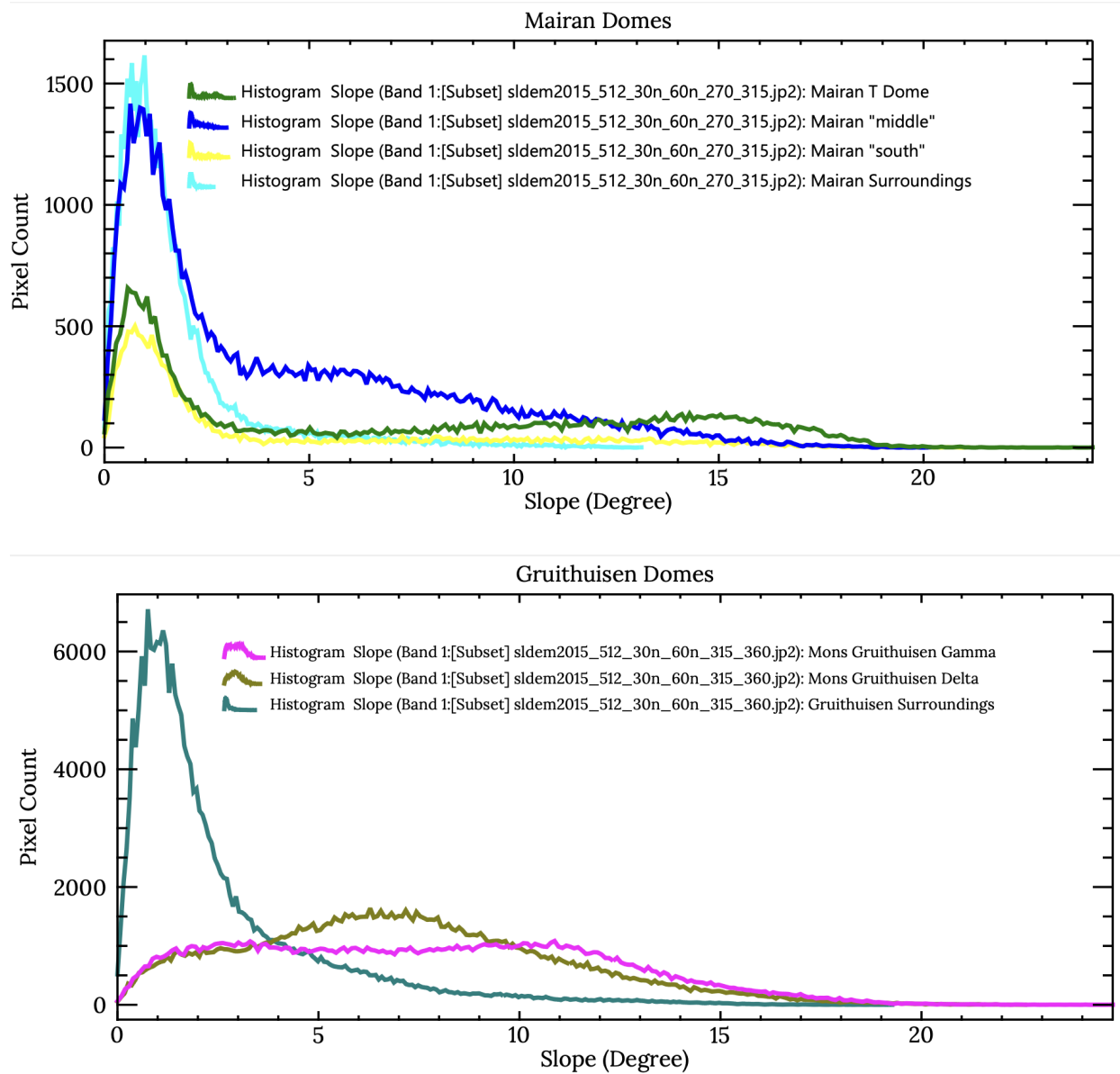


Fig. 5: Slope Distribution of ROIs for Gruithuisens and Mairans. They show distinct slope distributions in comparison to the surrounding controlled areas in which Mairans (upper) show relatively smaller mean slope values and Gruithuisens (down) appear to be steeper with a mean of higher slope values.

Results

CF-FeO anomaly	Min	Mean	Max	StdDev
Highlands Baseline	-0.050	-0.008	0.041	0.015
Gruithuisen Mare Baseline	-0.062	0.000	0.069	0.014
Mairan Mare Baseline	-0.039	0.000	0.049	0.012
Gruithuisen Gamma	-0.326	-0.062	0.055	0.045
Gruithuisen Delta	-0.399	-0.047	0.077	0.051
Mairan T	-0.227	-0.076	0.046	0.059
Mairan Middle	-0.268	-0.065	0.094	0.040
Mairan South	-0.186	-0.100	0.031	0.054
Mairan North	-0.096	-0.032	0.024	0.027
CF (μm)	Min	Mean	Max	StdDev
Highlands Baseline	8.16	8.19	8.23	0.013
Gruithuisen Mare Baseline	8.19	8.30	8.34	0.012
Mairan Mare Baseline	8.25	8.29	8.33	0.011
Gruithuisen Gamma	7.82	8.15	8.28	0.050
Gruithuisen Delta	7.76	8.18	8.31	0.063
Mairan T	8.00	8.12	8.25	0.058
Mairan Middle	7.99	8.15	8.26	0.035
Mairan South	8.02	8.10	8.21	0.055
Mairan North	8.13	8.22	8.29	0.038

Table 1: CF-FeO anomaly (top) and CF wavelength (bottom) statistics for the two Gruithuisen domes, four Mairan domes, and baseline terrain for comparison.

Statistical summaries (Table 1) revealed an average CF-FeO anomaly of -5.4% for the Gruithuisen domes and -6.8% for the Mairan domes while highlands and local mare were <1% anomalous. This indicates the Gruithuisen and Mairan domes trend toward proportionally higher silica content than is characteristic of typical lunar highlands and maria bulk mixtures. This is reflected quantitatively by the CF wavelength. The average CF wavelength across both sites were shifted to shorter wavelengths than their local terrain by 0.14 μm —which is larger than the 0.10 μm average CF difference between the lunar highlands and maria reported by Lucey et al. (2017) and observed by local terrain baselines. CF and CF-FeO both indicate an approximately uniform mean composition when comparing an average across all domes in each of the two sites, with the Mairan domes showing slightly

higher deviations from the expected composition. However, minimum and maximum values indicate a greater compositional range for the Gruithuisen domes compared to the Mairan domes. Despite its highest average mean, Gruithuisen Delta had our shortest measured CF at $7.76\mu\text{m}$, a surprising $0.4\mu\text{m}$ shorter than the highlands minimum. Standard deviations reveal more variability across the domes than baseline areas. Though the datasets are largely corrected to account for factors like illumination, space weathering, and slope, variability described by standard deviation could be a relict feature of the observation conditions. Thermal infrared data from LRO Diviner confirm the Gruithuisen and Mairan domes are relief features that exhibit higher than average silica proportions when compared with both local and global terrain. Though bulk compositions of the two sites exhibit similar spectral characteristics, CF and CF-FeO suggest a higher overall silica anomaly for the Mairan site. While Gruithuisen domes have a wider compositional range and the most extreme shifts toward shorter Christiansen Feature wavelengths.

The slope metrics (Fig. 6) indicate that both the Gruithuisen and Mairan edifices are topographically distinct from their surrounding plains, with dome ROIs exhibiting higher mean slopes and substantially longer high-slope tails than adjacent control terrains. Surrounding mare surfaces are dominated by very gentle slopes (mean $\approx 1.77^\circ$ at Mairan and $\approx 2.59^\circ$ at Gruithuisen), whereas the domes shift the slope distributions toward steeper values and reach much higher local extremes (maxima $\approx 20\text{--}25^\circ$ across all dome ROIs). All flank ROIs are markedly steeper than both the full-dome averages and the surroundings (flank means $\approx 8.83\text{--}10.55^\circ$ at Gruithuisen and $\approx 9.89\text{--}13.73^\circ$ at Mairan), demonstrating that dome steepness is concentrated in the cap-to-flank transition and outer flanks rather than being uniformly distributed. Notably, the steepest average flanks occur at Mairan T Dome ($\approx 13.73^\circ$), implying comparatively abrupt margins and/or more strongly preserved flanks, whereas Gruithuisen flanks are steep but less extreme in their mean values. As recent work suggests that silicic volcanism in the broader Gruithuisen region extends beyond the three principal domes (Kumari et al., 2024), integrating these slope constraints with CF distributions will strengthen our interpretations of silicic dome emplacement and modification relative to their surrounding mare surfaces.

Meanwhile, the Mairan slope histograms commonly retain a pronounced low-slope mode that resembles nearby mare, even while showing a distinct high-slope tail. This pattern is consistent with the geologic architecture inferred for the Mairan system: Boyce et al. (2018) describe Mairan Middle Dome as a composite silicic complex built from multiple small edifices and emphasize that basaltic lavas resurfaced and flooded the surrounding terrain during major volcanic episodes, while silica-rich eruptions were focused on the edifices and summit regions. Such a configuration naturally yields broad, low-gradient inter-edifice “platform” areas (many pixels at low slope) together with spatially limited steep margins and crater-wall segments (fewer pixels forming the high-slope tail). Likewise, independent remote-sensing studies indicate that Mairan T is embayed by younger mare deposits, implying that mare draping of dome toes and subsequent surface modification can further increase the fraction of gentle slopes within a dome ROI even where steep constructional fronts persist (Glotch et al., 2011).

Taken together, the Gruithuisen domes concentrate their highest gradients in narrow belts that outline the dome perimeter and locally emphasize lobate or scraped segments rather than showing uniformly deep topography. The Mairan domes genuinely have steep local slopes and still show a dominant shallow peak as they are embayed by mare basalts and composed of broad low-relief surfaces with steepness concentrated in limited, narrow zones.

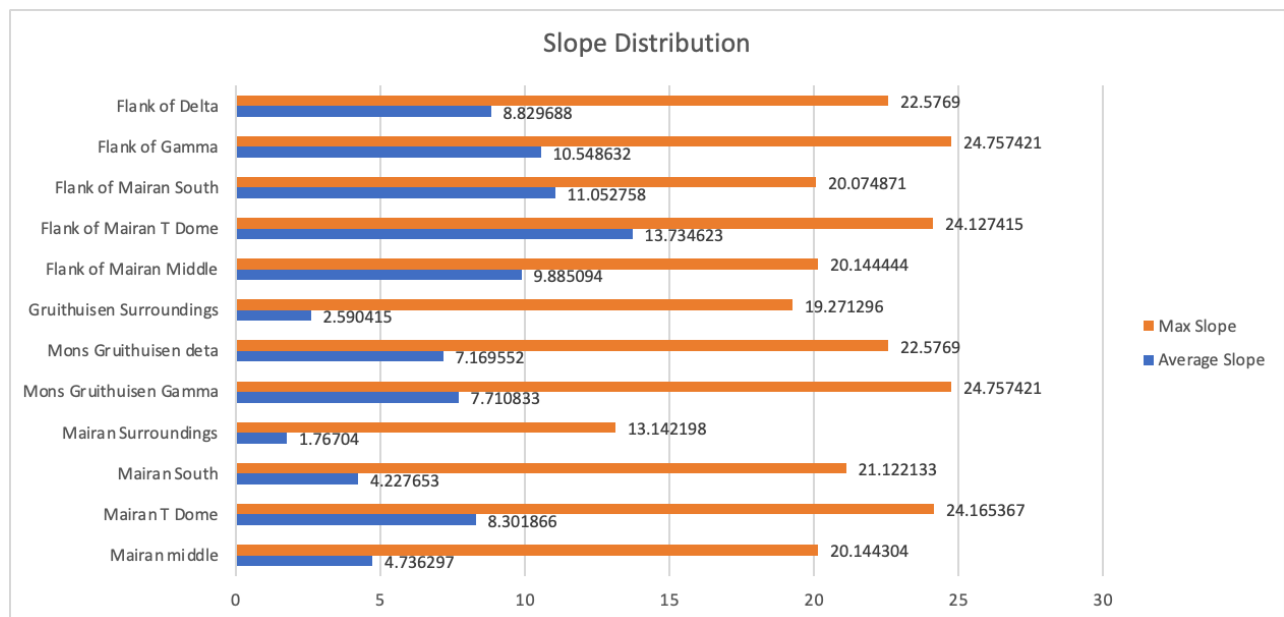


Fig. 6 compares mean slope (blue) and maximum slope (orange) for the Gruithuisen and Mairan domes, their mapped flank domains, and adjacent surrounding terrains. Flanks exhibit systematically higher average slopes (~8.8–13.7°) than the full-dome ROIs and far exceed surrounding mare means (~1.8–2.6°), while maximum slopes reach ~20–25° on domes/flanks but are lower in the surrounding terrains (notably ~13° at Mairan).

Discussion

Terrestrial rhyolitic lava domes provide a useful physical analogue for interpreting why the Gruithuisen and Mairan edifices are unusual on the Moon. On Earth, rhyolite is silica-rich magma whose high viscosity favors emplacement as thick, blocky flows and steep-sided domes, rather than low-profile sheet flows typical of basalt. Dome morphologies and flank steepness are commonly understood as outcomes of viscoplastic flow, where magma yield strength, effusion rate, and cooling-controlled crust formation govern whether domes grow as short, steep piles, elongate into coulees, or develop lobate margins and breakout-like features. Experimental and field-based syntheses further illustrate that many rhyolitic domes are polygenetic, assembled through multiple eruptive pulses and shifting vents, which can produce composite edifices marked by overlapping lobes and internal breaks-in-slope (Fink and Griffiths, 1998). In addition, lava domes are often mechanically unstable, with oversteepened margins and talus-prone flanks that can fail episodically; although lunar domes do not produce pyroclastic density currents in the terrestrial sense, the same geometric tendency toward margin steepening and gravitational adjustment stimulate explicitly separating cap, margin, and flank slope statistics when comparing edifices to their surrounding plains. Dome-like morphologies and steepened flanks are typically associated with high-viscosity, silica-rich magmas on Earth, yet the Moon's volcanic record is dominated by low-viscosity basalts. Therefore, documenting whether the Gruithuisen and Mairan domes concentrate steep slopes preferentially on flanks and whether those zones align with silica-rich orbital proxies) is a direct test of constructional emplacement versus later modification.

Trends emerge when we compare our findings from LOLA slope with Diviner composition. Of the domes, Gruithuisen Delta had the highest average CF wavelength at 8.18 μ m, suggesting that it

has the least silicic and therefore lowest viscosity flow. This aligns with it having the least steep average flanks and lowest mean overall dome slope. By contrast, the Mairan domes generally display shorter CF wavelengths, indicating more silica-rich compositions; our slope results support this inference by documenting steeper flank gradients that are consistent with higher-viscosity emplacement. Morphological-compositional linkages also appear within the Mairan group. Mairan South, the most silicic dome identified by CF placement, exhibited slope characteristics suggesting a similar emplacement method to rhyolitic constructions. This dome has the second steepest flank yet its maximum measured slope comes from elsewhere on the dome, implying localized steep segments superposed on broader low-gradient surfaces. Further, despite having the shortest CF wavelength and smallest range, and therefore high viscosity fluid, Mairan South has the lowest overall slope. These characteristics resemble rhyolitic composite constructions, where several silicic heterogeneous lobes are likely the reason for a short CF position accompanied by an overall low average slope as the edifice-wide slope remains muted due to extensive gently sloping cap and inter-lobe areas. Consequently, subdividing ROIs into summit, flank, and margin subdomains is essential for isolating the most constructionally diagnostic and potentially most silica-enriched parts of each dome when comparing topography to Diviner-derived compositional proxies.

Conclusion

The Gruithuisen and Mairan domes are examples of anomalous silicic volcanism on the moon as evidenced by their spectral characteristics in the thermal IR and slope morphology. For the Gruithuisen domes, relief is expressed primarily as broad, gently sloping summit domains that transition outward into distinct belts to flank steepening and sharp margin slope breaks. This organized geometry offers a clear basis for testing whether CF-FeO anomalies preferentially coincide with particular morphologic domains versus being influenced by topographic roughness. Among the Mairan domes, slope distributions instead reflect a patchier arrangement in which localized high-gradient segments are superimposed on low-slope terrain. Such a pattern supports the inference that mare embayment and resurfacing have expanded the area of mare-like gentle slopes within dome footprints. Our comparison between Christiansen Feature and dome-scale elevation characteristics

revealed a relationship between typically high silica proportions and increased slope angles. Unlike typical lunar flood basalts, these silicic domes exhibit characteristics similar to terrestrial rhyolitic volcanism through steep-sided flanks and heterogeneous slope profiles caused by high viscosity composite emplacement. Silicic magma on Earth typically form near areas of mantle upwelling or subduction zones (Bachmann & Bergantz 2008), so identifying similar dome characteristics on the Moon in the absence of water and plate tectonics frames future research for the geologic history of lunar silicic volcanism.

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